
Real-World Mini CTI
Investigation Lab
OSINT Analysis • Infrastructure Pivoting • ATT&CK Mapping
VirusTotal
Analyze. Detect. Respond.
d41d8cd98f00b204e9800998ecf8427e
Detections 0 / 72
Community Score 0
Last Analysis 2025-08-30 12:34:56 UTC
ALIENVAULT
OTX
Open Threat Exchange
Indicators
Reports
Malware
Campaigns
Related Feeds
01001000 0110
01010111 01100
01101111 01100
d41d8cd98f00b204e980
0998ecf8427e
e3b0c44298fc1c149afbf
4c8996fb92427ae41e464
MITRE ATT&CK
Tactics | Techniques
Initial Access
Execution
Persistence
Privilege Escalation
Defense Evasion
Credential Access
Discovery
Lateral Movement
Collection
Command and Control
Exfiltration
Impact

Abdul Wahid

Executive Summary

This investigation examined three publicly available malware hashes using free OSINT threat intelligence tools (VirusTotal, AlienVault OTX, AbuseIPDB, and urlscan.io).

Hash-1 showed clear Trojan/Stealer/Evader behavior with multiple persistence and injection techniques.

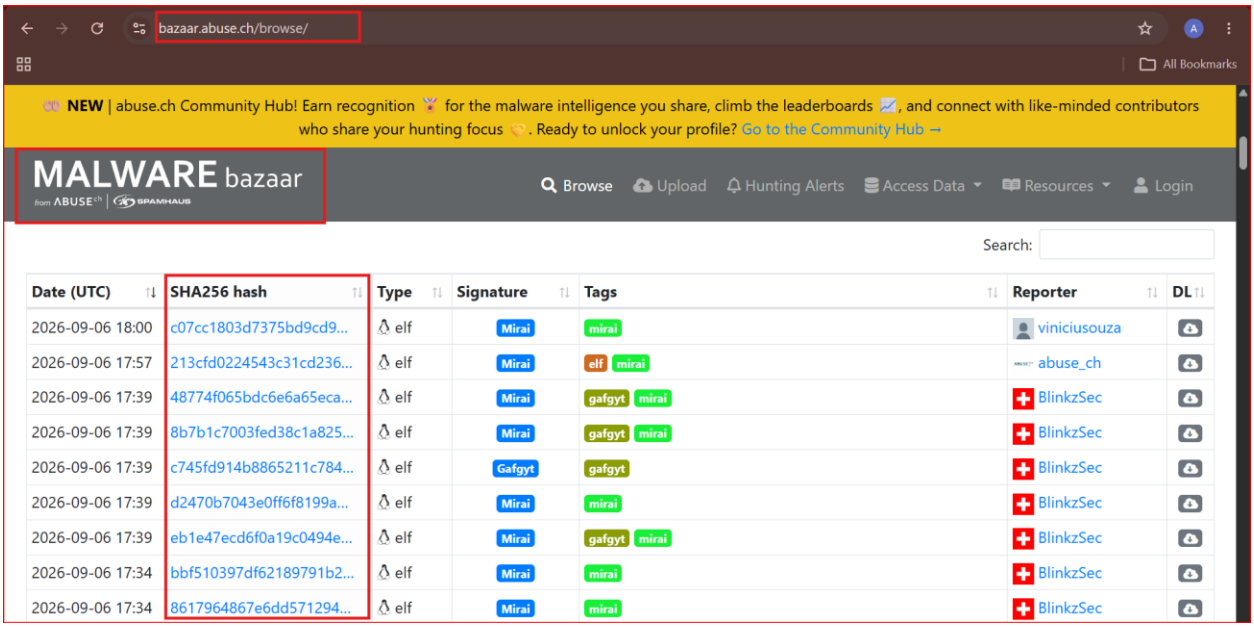
Hash-2 was confirmed as AgentTesla through community pulses.

Hash-3 provided useful related infrastructure, including a suspicious Russian-hosted panel (103.101.85.26) that resembles a malware C2/gate.

Overall, the samples demonstrate typical information-stealer tactics. No active exploitation against the analyst's environment was observed, but the indicators should be blocked and monitored.

Requirement

Before starting the lab save a hash of malware from Malware Bazaar



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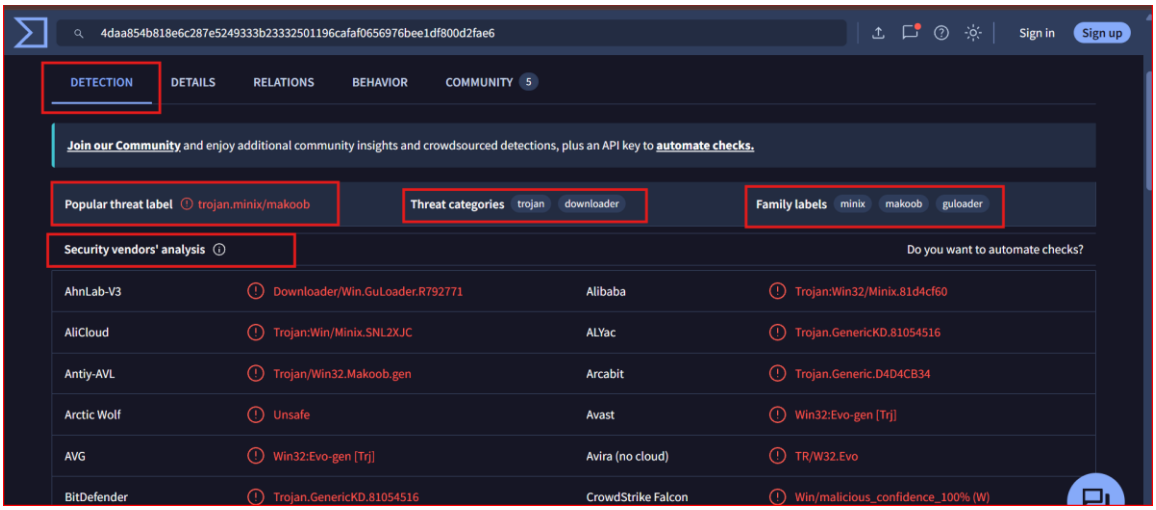
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VirusTotal Investigation Hash-1

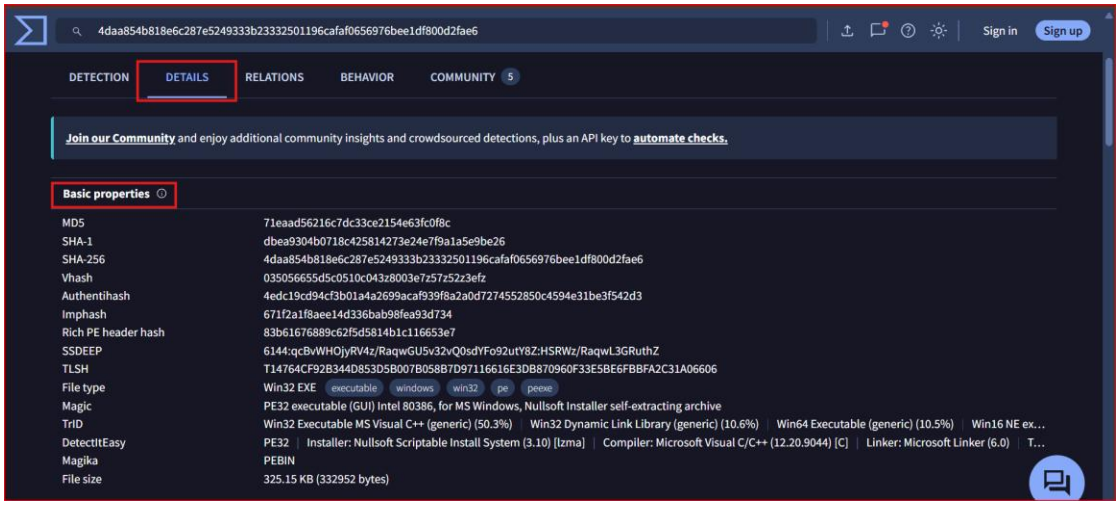
Tool: VirusTotal
Indicator (SHA-256):
4daa854b818e6c287e5249333b23332501196cafaf0656976bee1df800d2fae6

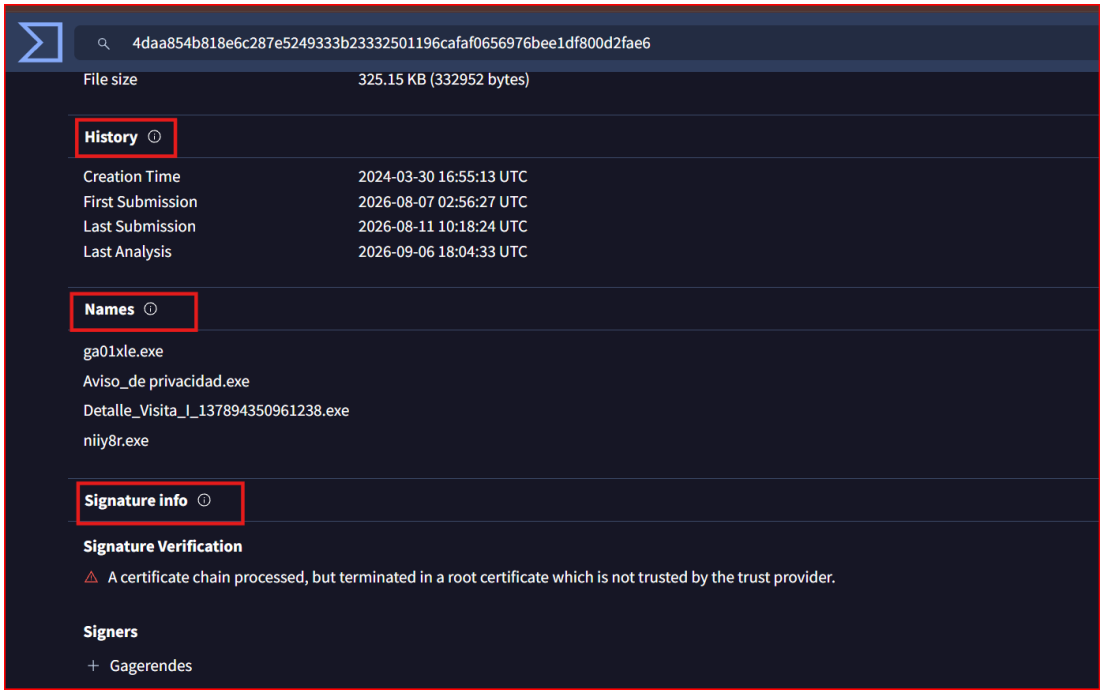
Steps:

1. Go to https://www.virustotal.com and search the hash.
2. **Detection Tab:**
    - Record Detection Ratio (e.g. 51/70)
    - Note popular threat label / malware family
    - Note threat categories
    - List some security vendors that detected it

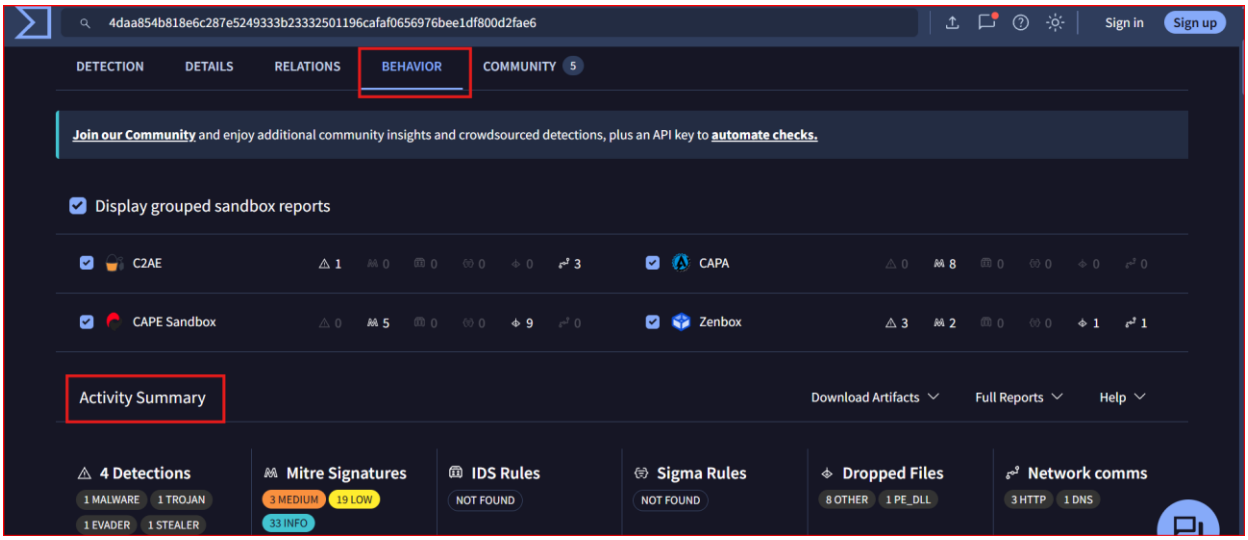


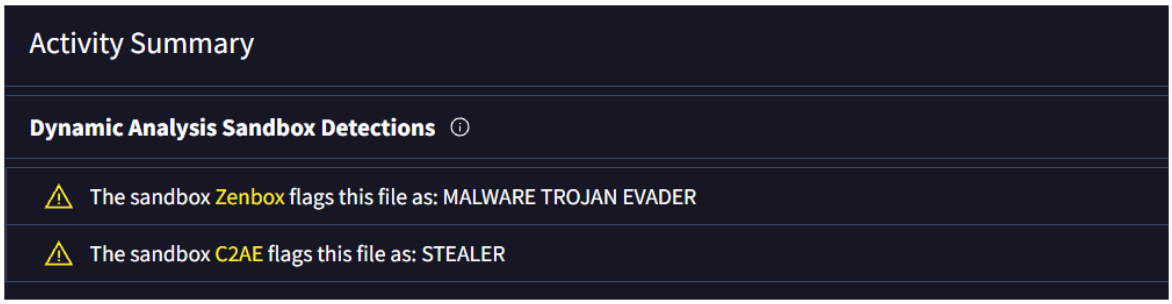
3. **Details Tab:**
    - Basic Properties
    - First Submission date
    - Last Analysis date
    - File names
    - Signature information



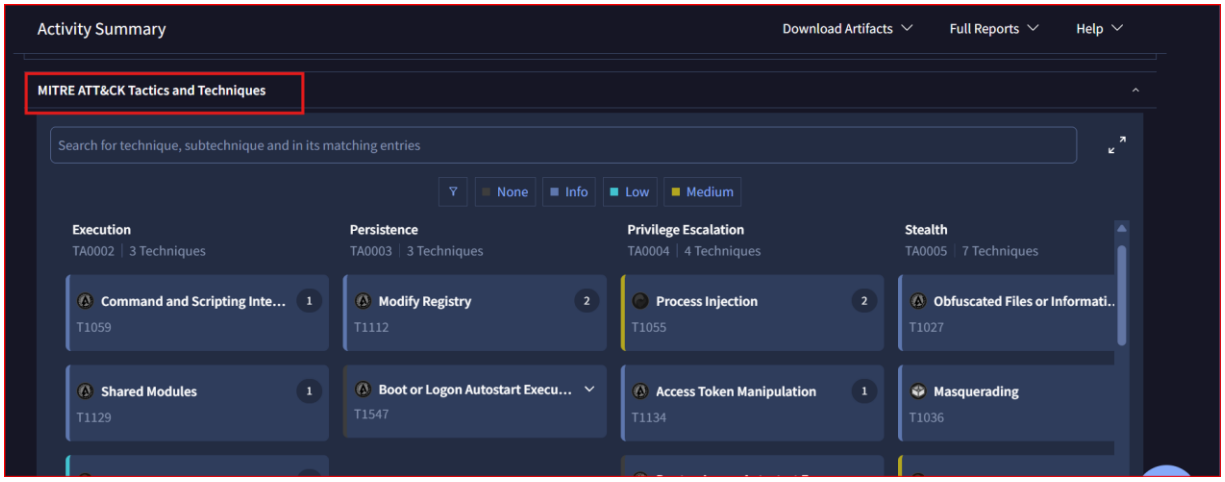


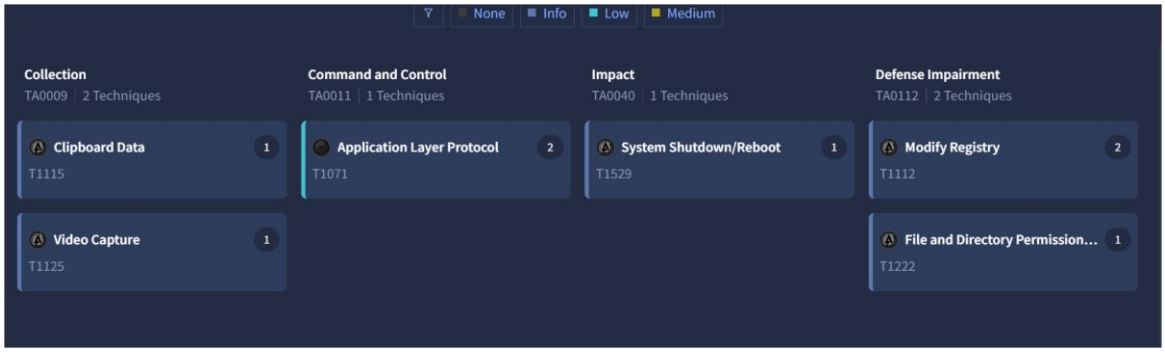
4. **Behavior Tab:**
    - Read Activity Summary
    - Note sandbox detections (e.g. Trojan, Stealer, Evader)

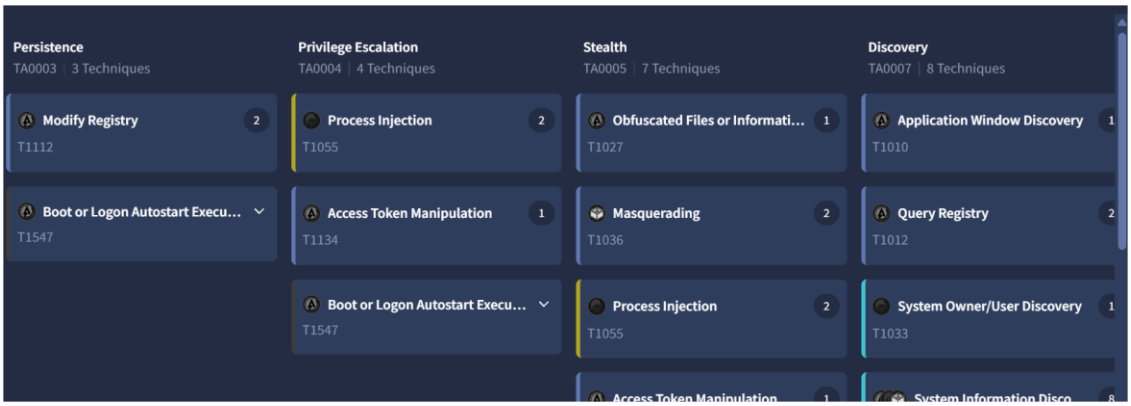




5 Analyze MITRE ATT&CK techniques shown





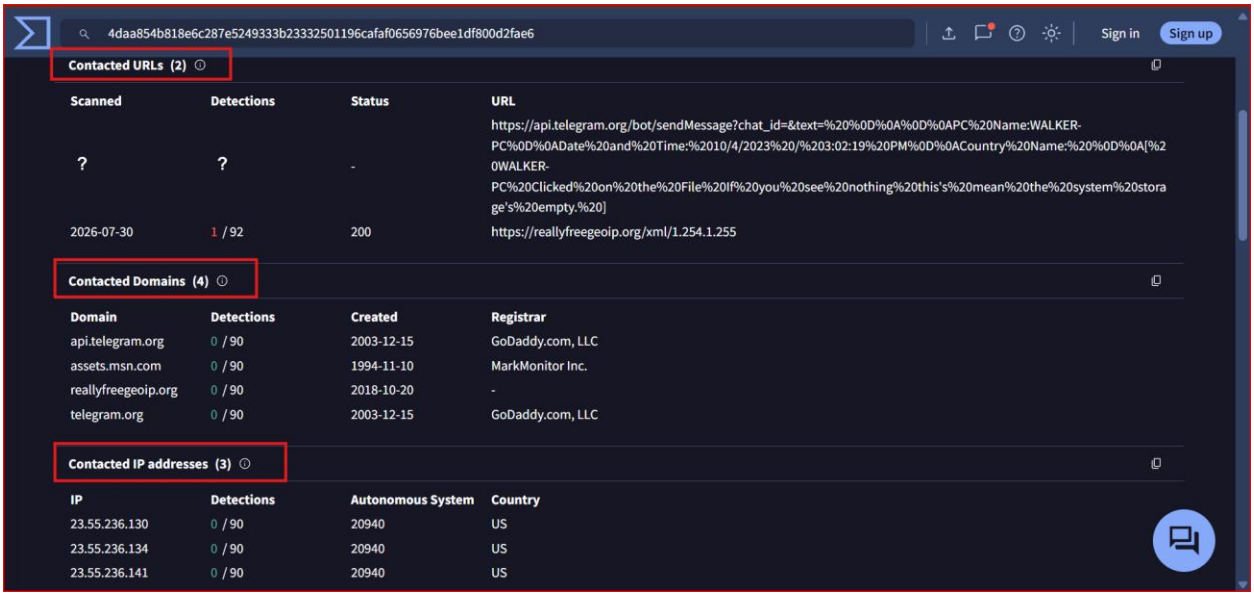


6 **Relations Tab:**

a. Check Contacted Domains

b. Check Contacted IP addresses

c. Check Contacted URLs

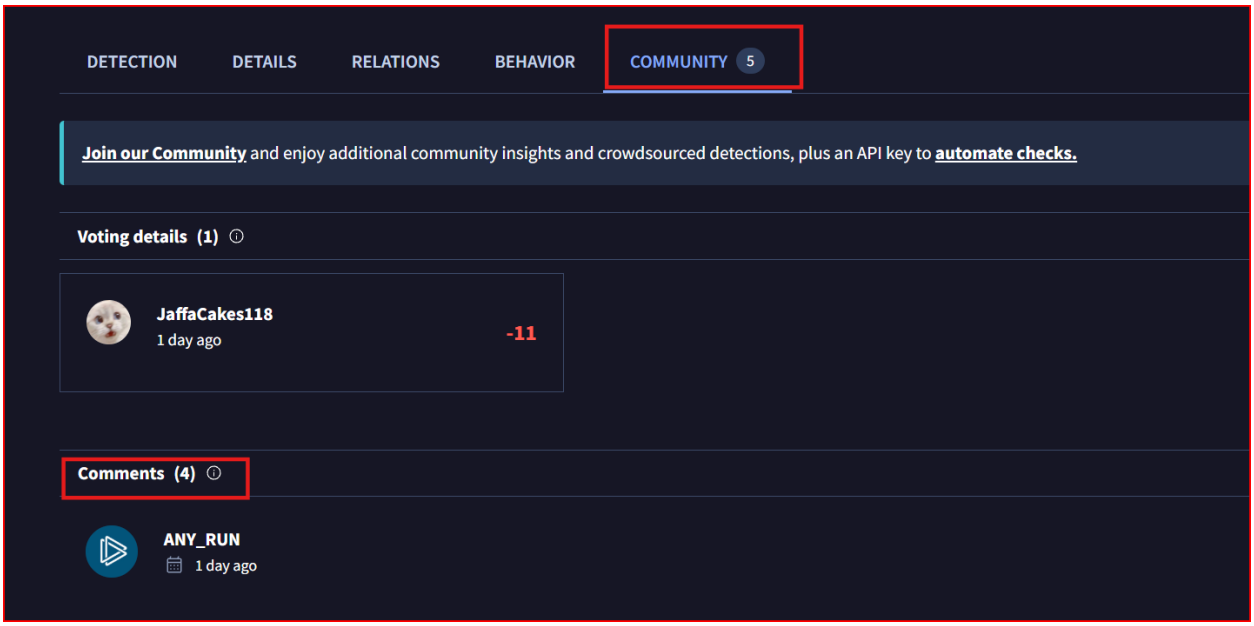


d. Review the Graph Summary



7 Community Tab:

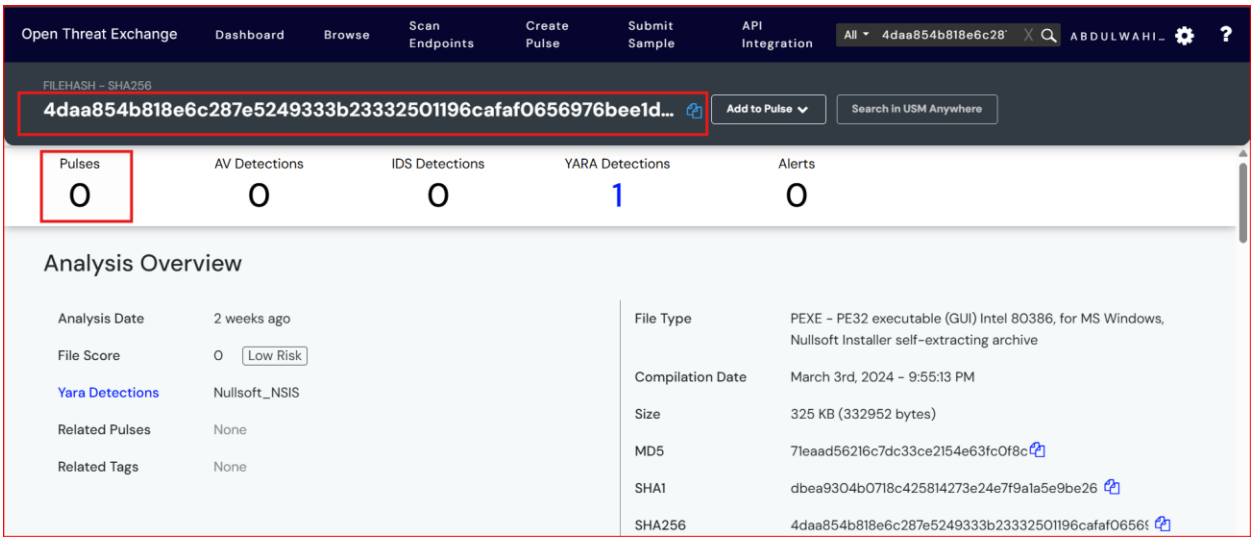
a. Check any useful comments or votes



AlienVault OTX Investigation Hash-1

Tool: AlienVault OTX
Indicator (SHA-256):
4daa854b818e6c287e5249333b23332501196cafaf0656976bee1df800d2fae6

1. Open your browser and go to: https://otx.alienvault.com
2. If you don't have an account, create a free account (required to see full details).
3. Paste the SHA-256 hash into the search bar and search.
4. On the results page, note the following:
    - **Pulse Count** (how many pulses mention this hash)

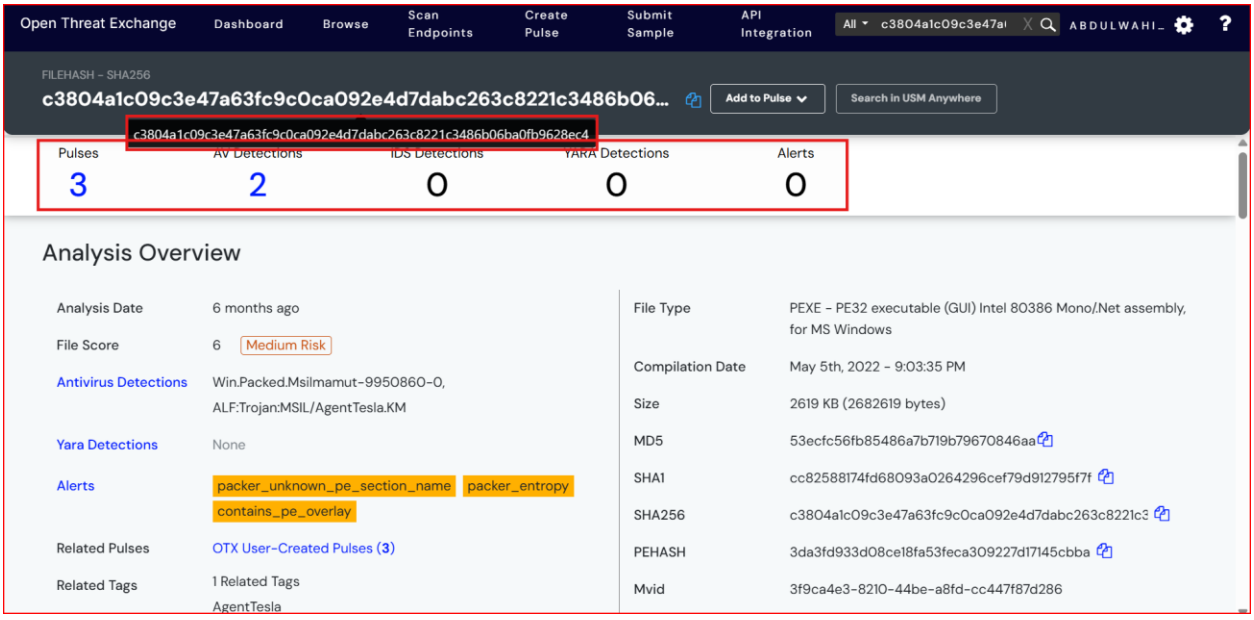


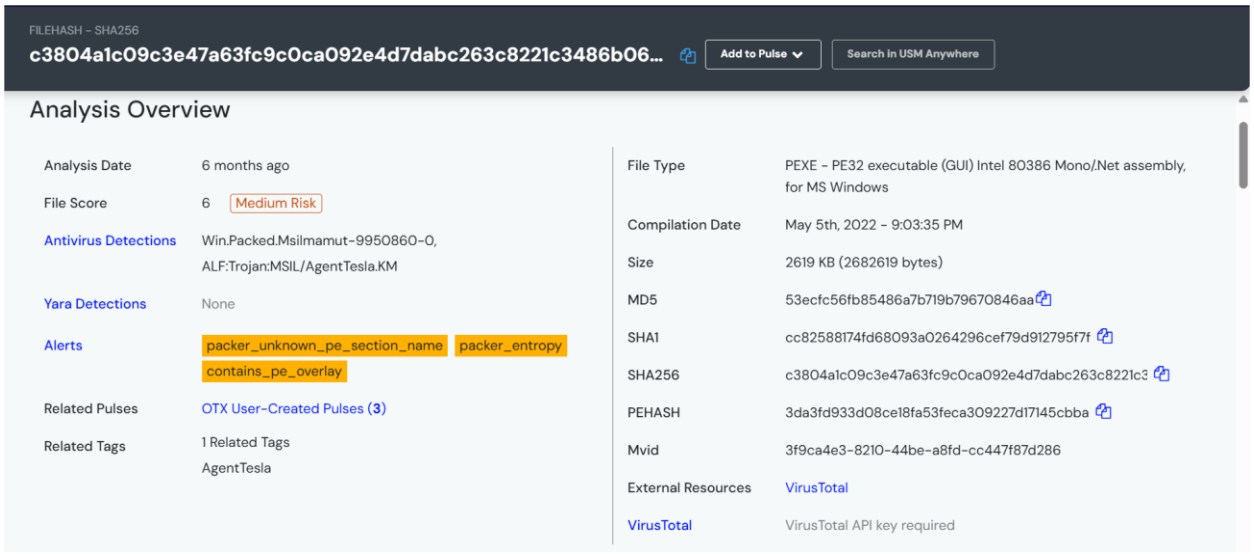
As the hash we selected has no pulses & we detect a single YARA rule here
Let's analyze another hash on Alien Vault OTx

AlienVault OTX Investigation Hash-2

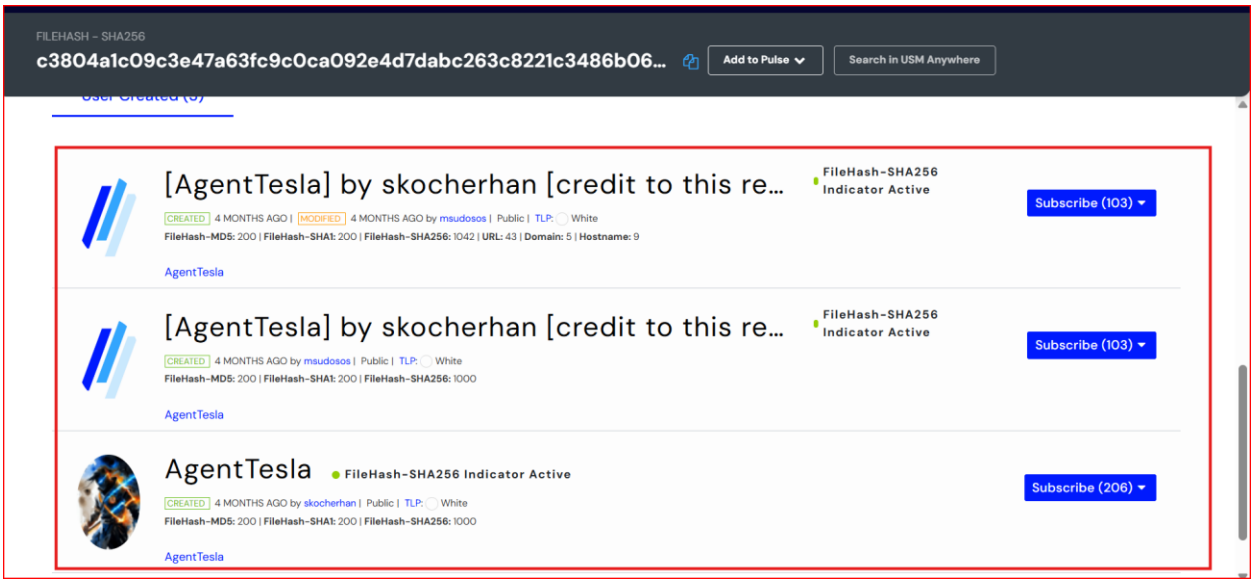
Indicator (SHA-256):
c3804a1c09c3e47a63fc9c0ca092e4d7dabc263c8221c3486b06ba0fb9628ec4

1. Paste the SHA-256 hash into the search bar and search.

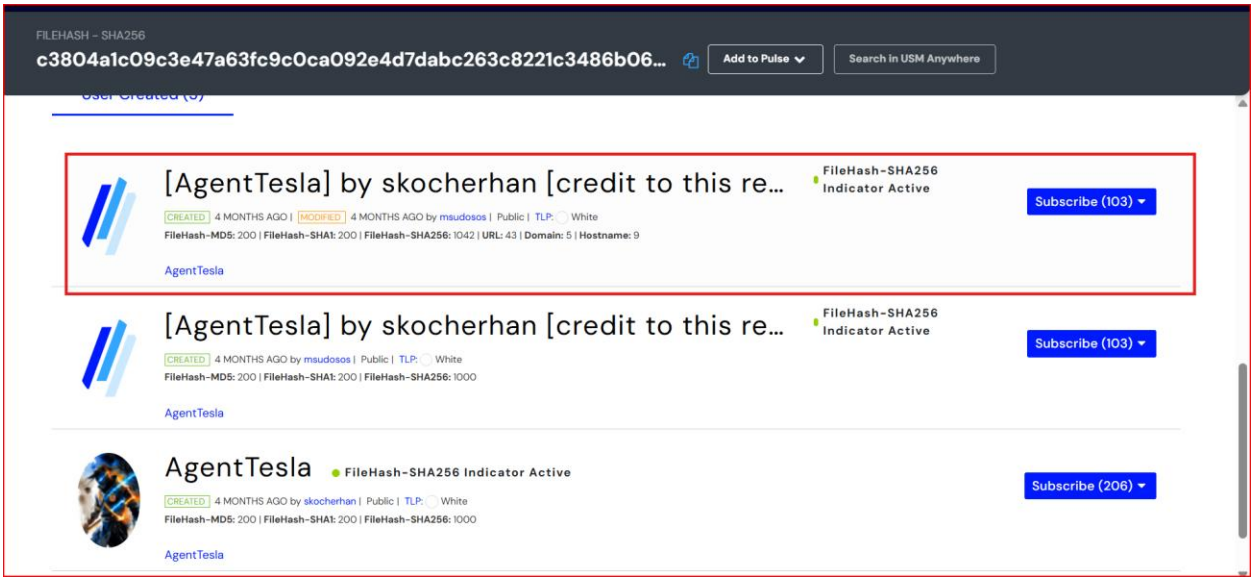




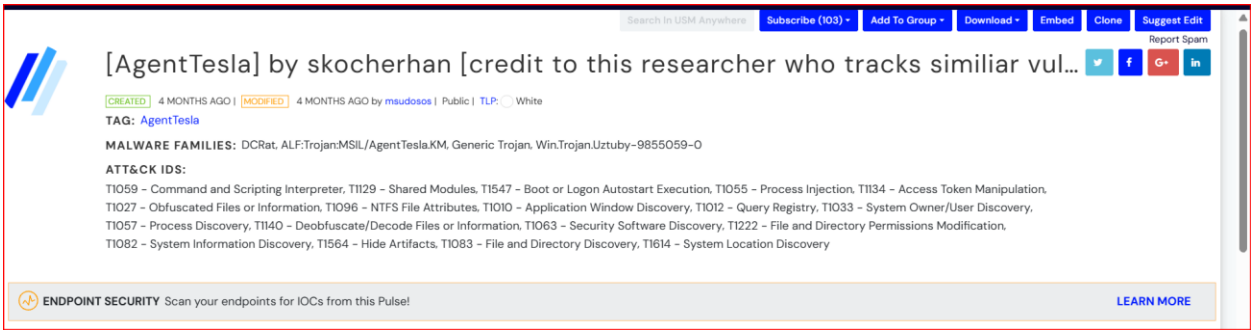
2. On the results page, note the following:
    - **Pulse Count** (how many pulses mention this hash)

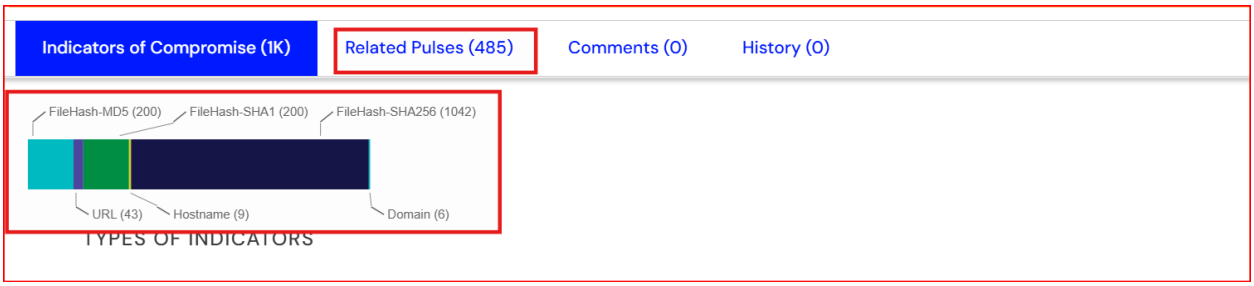


    - Open the most relevant **Pulses** (community threat reports)

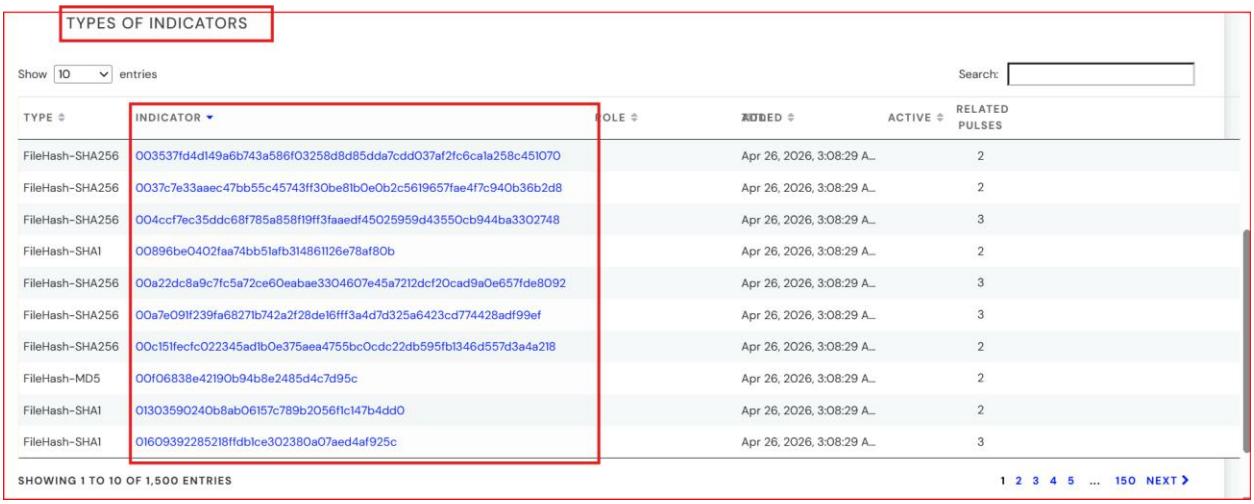


Pulse information





3. Check the **Indicators** section related to this hash.



Antivirus detection

Antivirus Detections

VENDOR	FINDING	NOTES
CLAMAV	Win.Packed.Msilmamut-9950860-0	Malware infection
Ms Defender	ALF:Trojan:MSIL/AgentTesla.KM	Malware infection

Files Matching Antivirus Detection

Number of samples

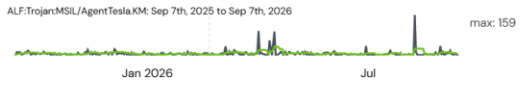
56K

Copy 1000 most recent hashes

Hide Samples

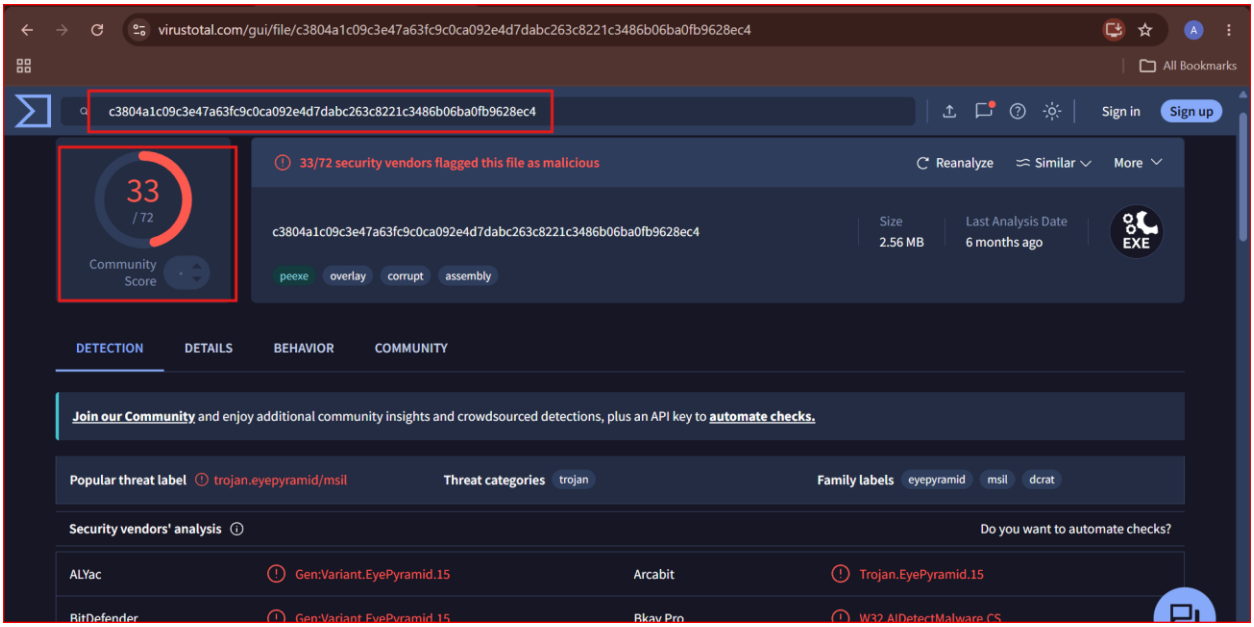
Last Sample Analysis

2 hours ago



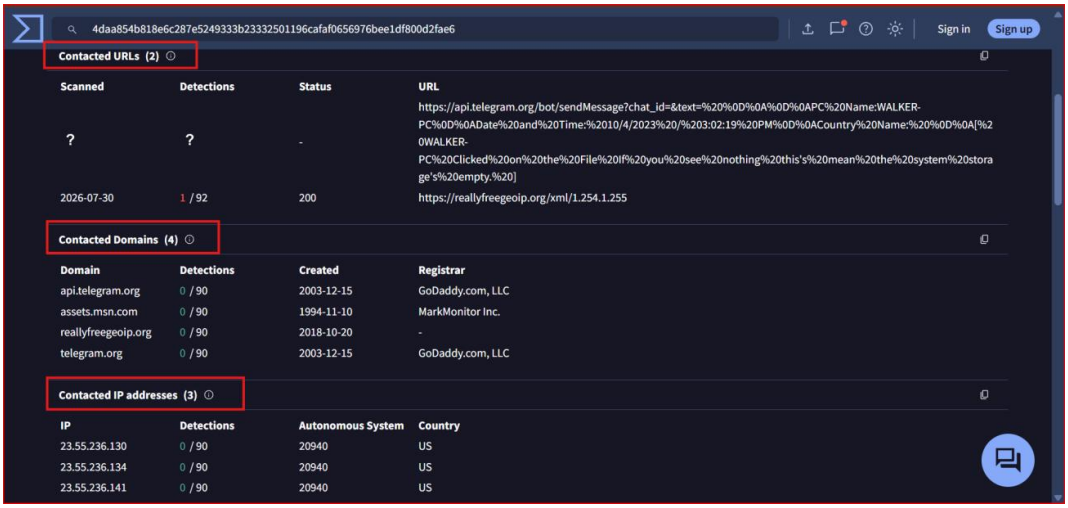
Virustotal investigation Hash-2 Verification

Checking the Hash does it flag in virustotal as malicious or not



Cross-Validation & Pivoting (Hash-1)

1. **Go back to your VirusTotal results**
2. Open the **Relations** tab.
3. From **Contacted Domains** or **Contacted IP addresses**, copy **2–3 indicators** (choose ones that look suspicious).



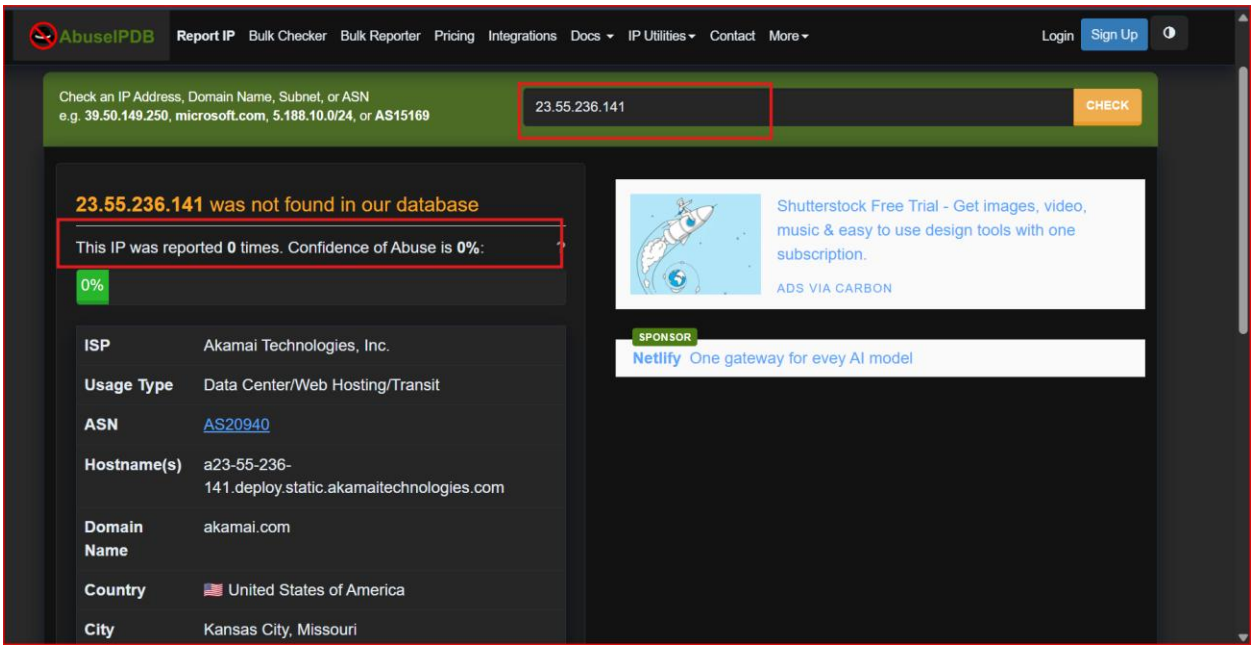
Check each selected indicator on the following free tools:

AbuseIPDB (for IP addresses)

Contacted IP addresses (3)

IP	Detections	Autonomous System	Country
23.55.236.130	0 / 90	20940	US
23.55.236.134	0 / 90	20940	US
23.55.236.141	0 / 90	20940	US

- Go to: https://www.abuseipdb.com
- Search the IP

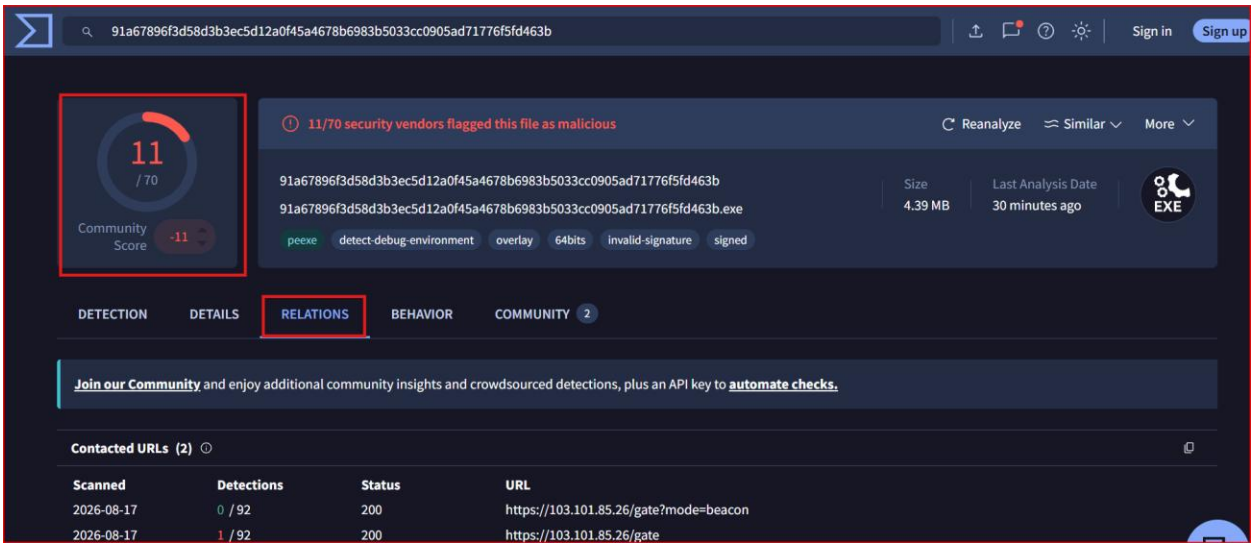


IMPORTANT

As these IP's aren't reported we will use **Hash-3** to continue analyze to get clear results

Hash-3 -> 91a67896f3d58d3b3ec5d12a0f45a4678b6983b5033cc0905ad71776f5fd463b

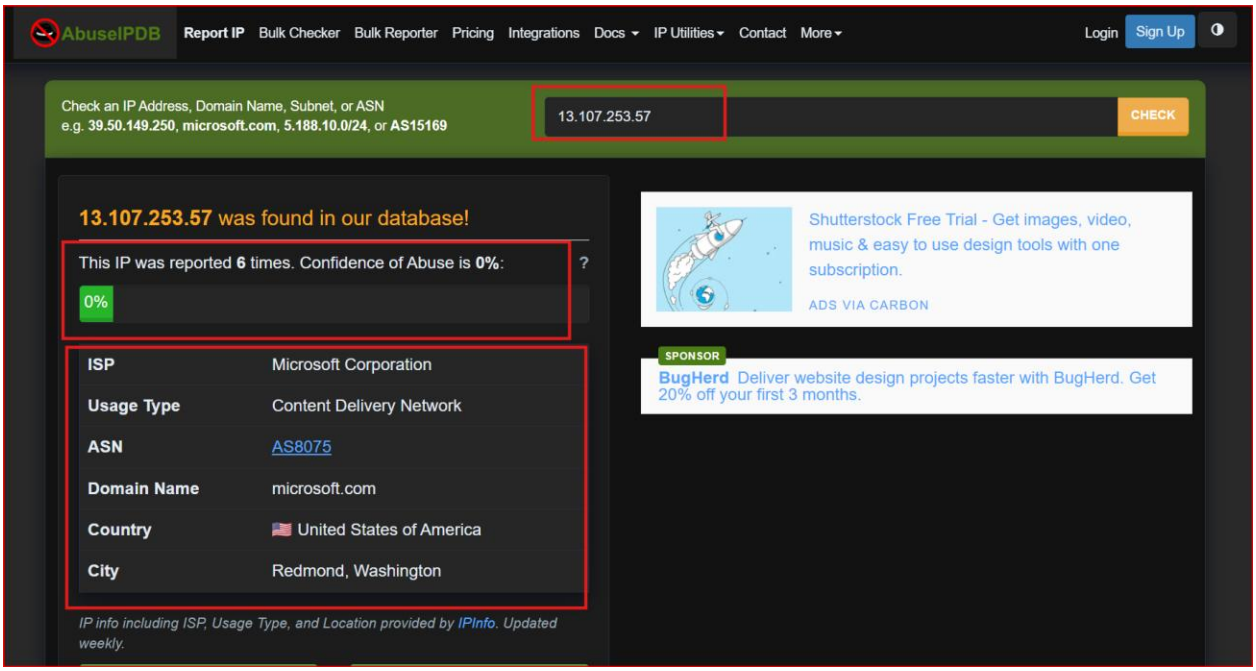
Hash-3 (Relation Check on Virus total)



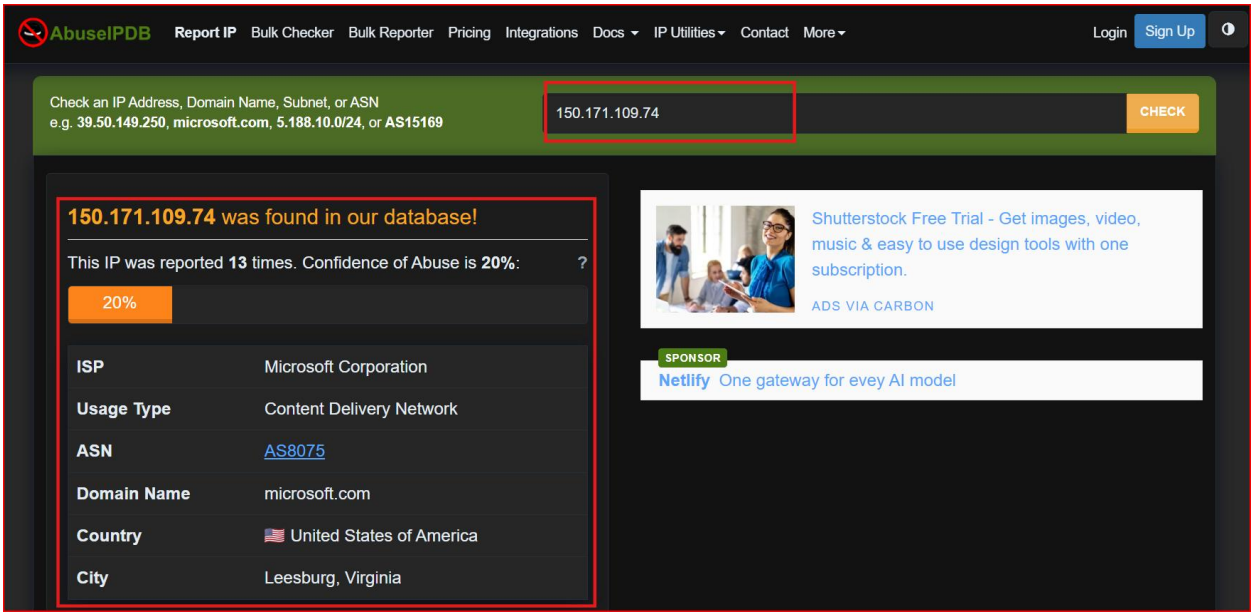
Search for the IP in AbuseIPDB



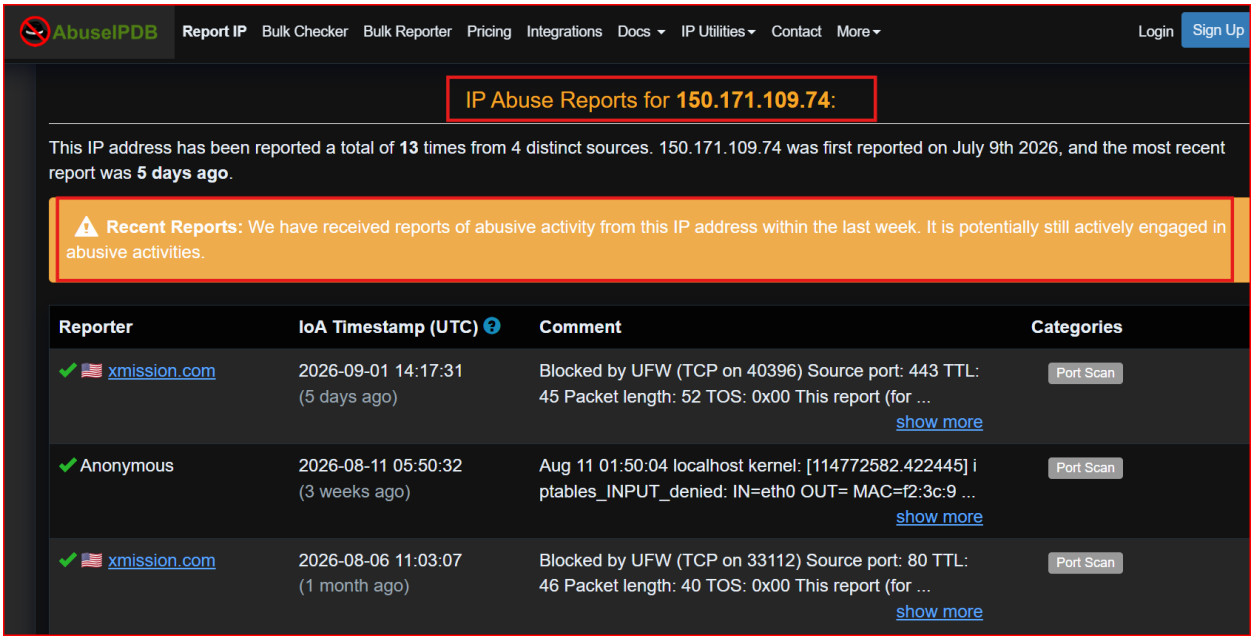
Note: Confidence Score, Number of reports, ISP, Country, and any attack types



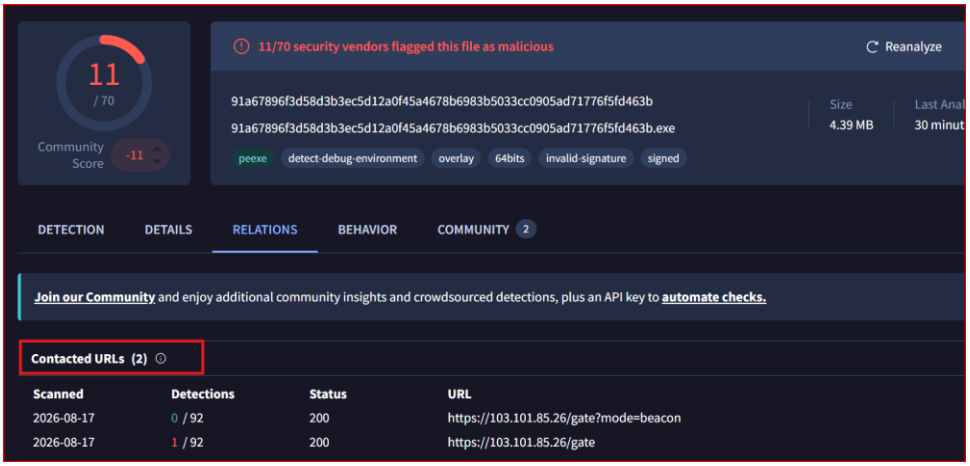
Another IP Checking



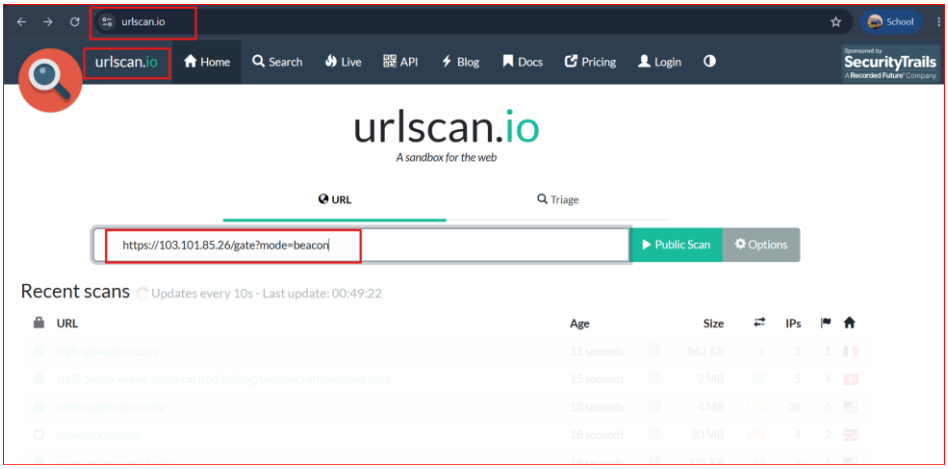
IP is still engaging



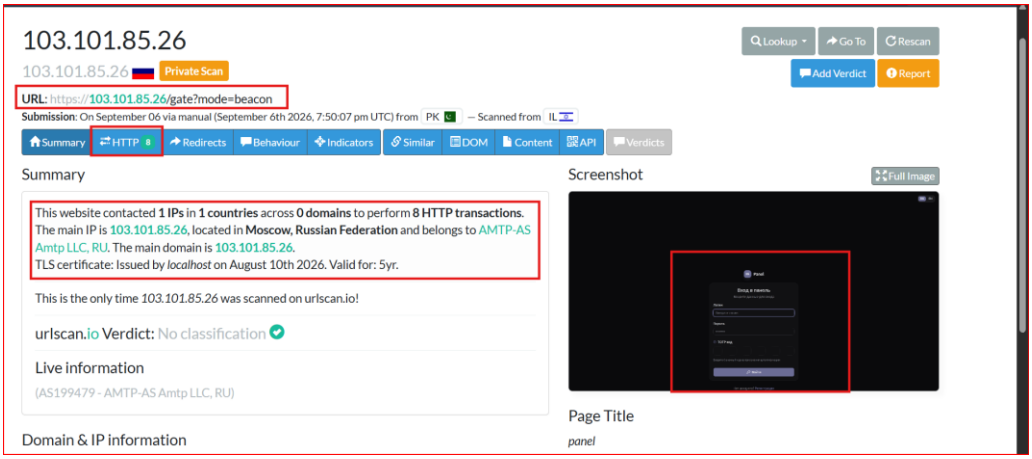
urlscan.io (for Domains / URLs)

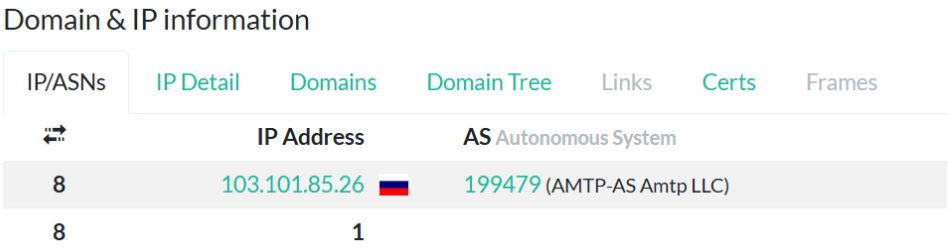


- Go to: https://urlscan.io
- Search the domain

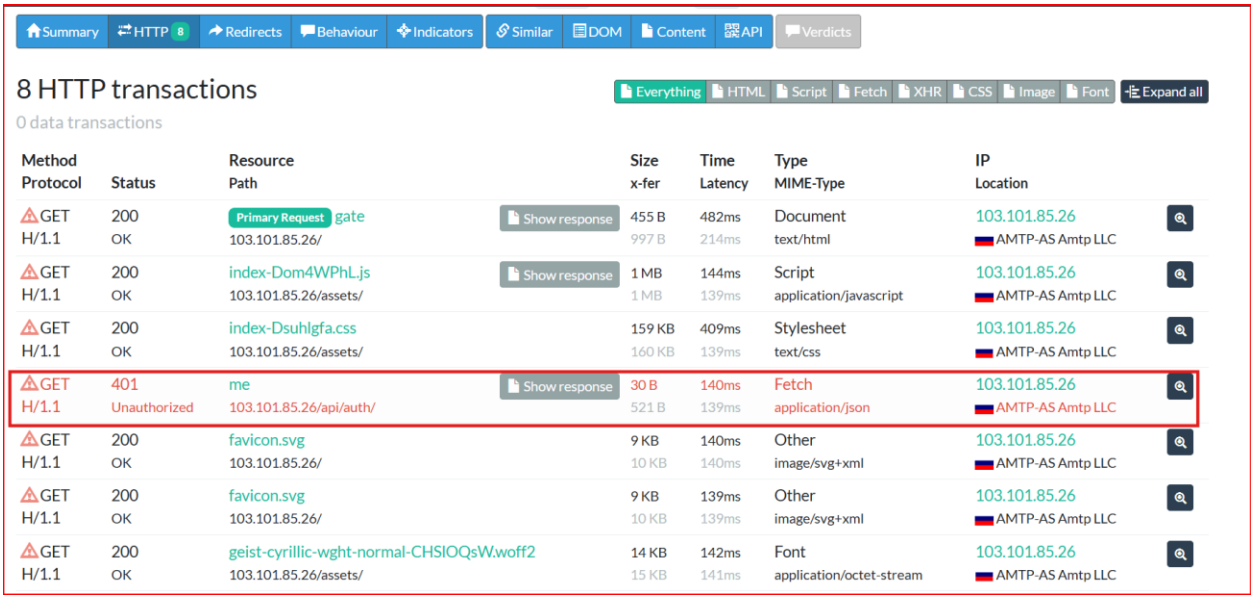


Note: Malicious score (if any), technologies, related IPs, screenshots of the page





This IP + URL combination is suspicious and commonly associated with malware command-and-control (C2) panels, especially with stealers. Even though urlscan has not classified it as malicious yet, the path /gate?mode=beacon + Russian hosting + login panel makes it a strong related indicator.

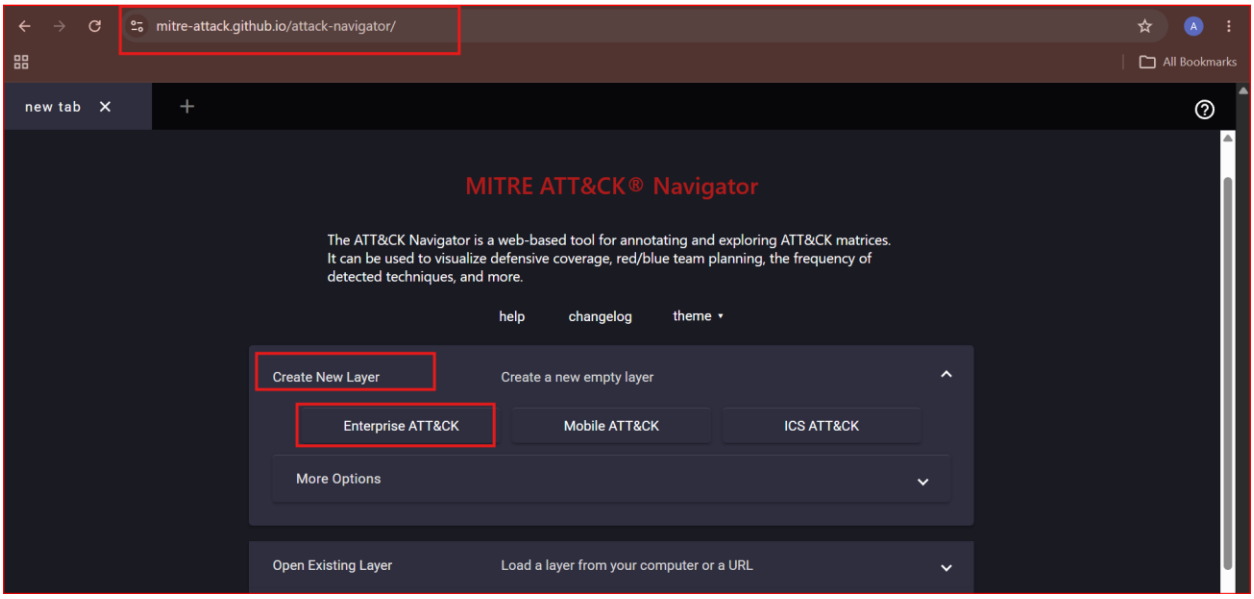


MITRE ATT&CK Navigator Mapping

Tool: MITRE ATT&CK Navigator
Link: https://mitre-attack.github.io/attack-navigator/

Clean Steps:

1. Open the ATT&CK Navigator link above.
2. Create a new layer (or use the default Enterprise layer).



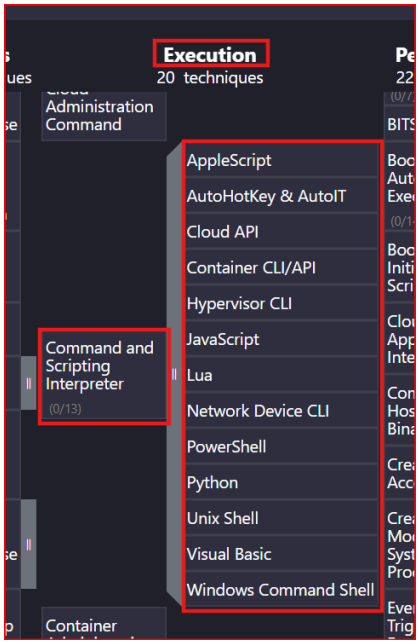
3. Based on what you found so far (especially from VirusTotal Behavior tab and OTX pulses for AgentTesla), **highlight at least 5–6 techniques**.

Common techniques you should consider mapping (from your previous results):

1. Command and Scripting Interpreter (T1059)

The malware uses scripting or command-line tools to run its code.

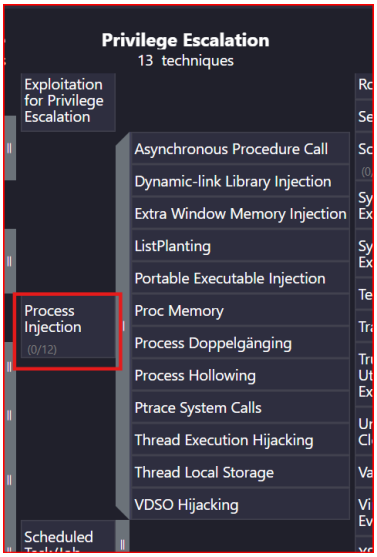
This matches the installer-style behavior seen in the VirusTotal analysis of Hash-1.



2. Process Injection (T1055)

It injects its code into other running processes to hide and stay active.

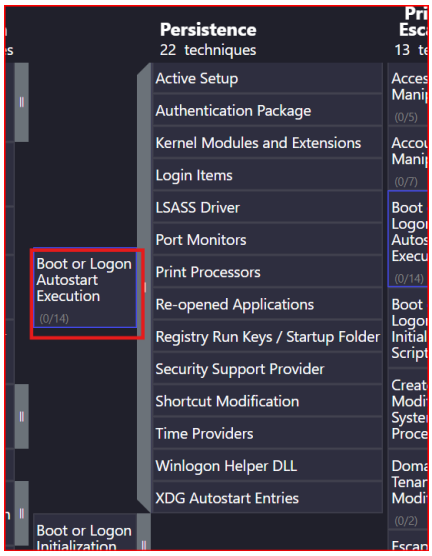
VirusTotal Behavior tab clearly flagged this technique for Hash-1.



3. Boot or Logon Autostart Execution (T1547)

The malware sets itself to start automatically when the computer boots or the user logs in.

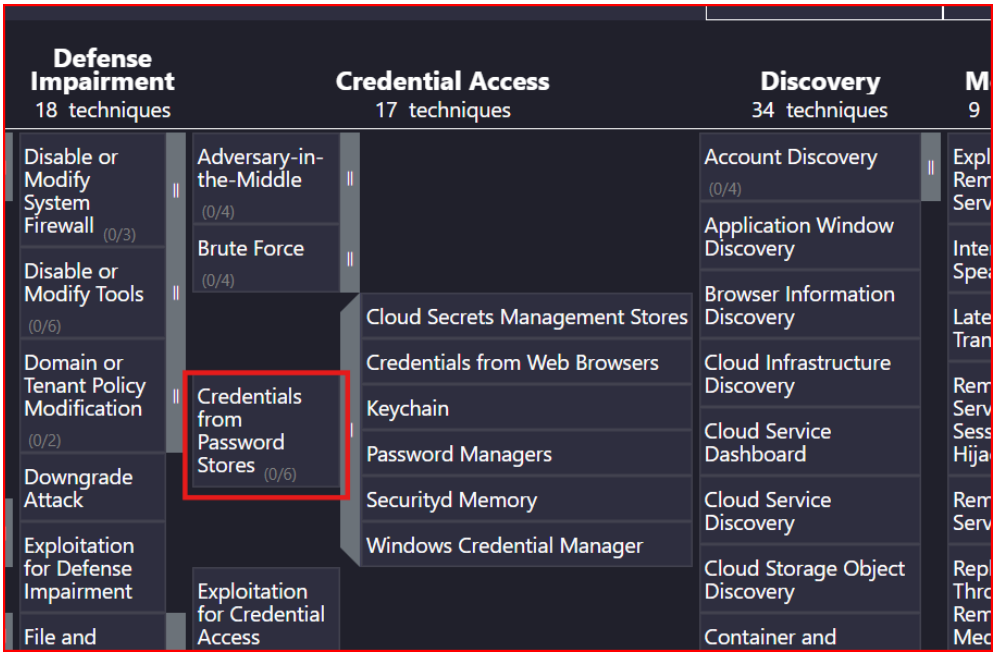
This was observed in the Persistence techniques shown for Hash-1.



4. Credentials from Password Stores (T1555)

It tries to steal saved passwords from browsers or system password stores.

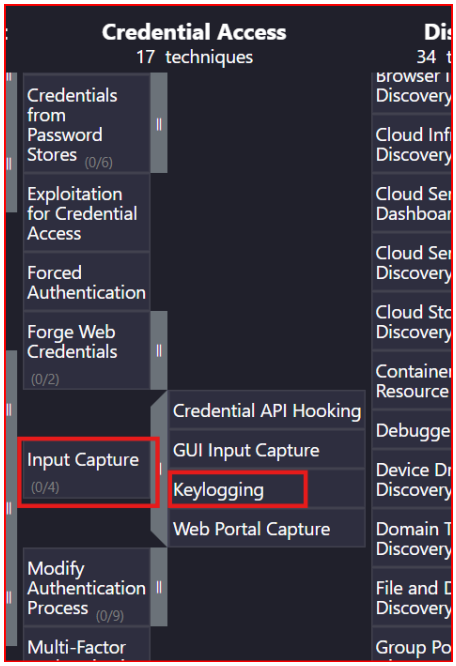
This fits the stealer-type detections (Trojan/Stealer) reported for Hash-1.



5. Input Capture (T1056) – Keylogging

It records what the user types on the keyboard.

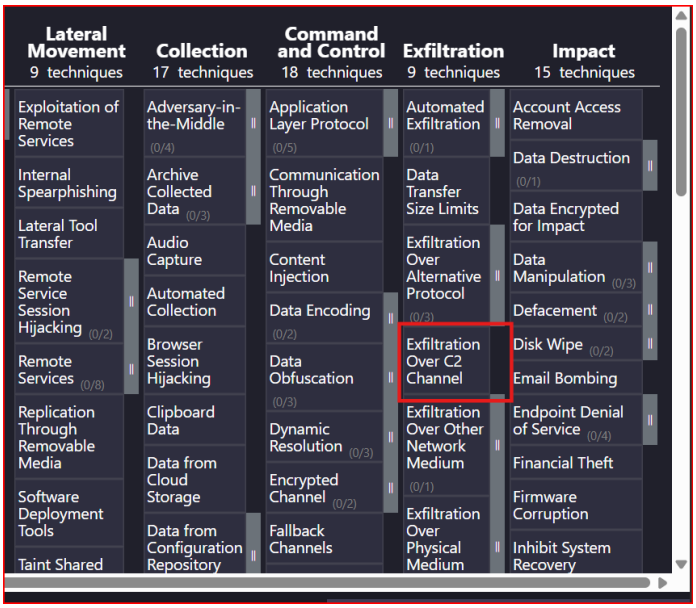
Common in stealer malware and consistent with the sandbox detections on Hash-1.



6. Application Layer Protocol (T1071)

The malware talks to its command server using normal web or application protocols.

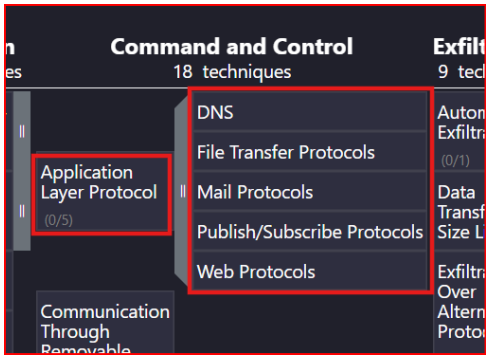
This matches the network communication patterns seen in the Relations tab of Hash-1.



7. Exfiltration Over C2 Channel (T1041)

Stolen data is sent back to the attacker through the same communication channel.

This is typical for the stealer behavior flagged on Hash-1.



Recommended Actions

Hash-1 (4daa854b818e6c287e5249333b23332501196cafaf0656976bee1df800d2fae6)

- Block the hash in endpoint protection / EDR.
- Hunt for the observed file names (gato1ole.exe, Avisos_de_privacidad.exe, etc.) across the environment.
- Monitor for Process Injection and Boot/Logon Autostart techniques associated with this sample.

Hash-2 (c3804a1c09c3e47a63fc9c0ca092e4d7dabc263c8221c3486b06ba0fb9628ec4)

- Treat as confirmed AgentTesla. Block the hash immediately.
- Search historical logs for any execution of this hash or related AgentTesla indicators from the OTX pulses.
- Alert on any outbound connections matching the C2 patterns linked to this family.

Hash-3 (91a67896f3d58d3b3ec5d12a0f45a4678b6983b5033cc0905ad71776f5fd463b)

- Block the hash.
- Especially block and monitor the related IP 103.101.85.26 and the URL path /gate?mode=beacon (strong C2/panel indicator).
- Add the Russian hosting ASN (AMTP-AS) to heightened monitoring if it appears in proxy/firewall logs.

Confidence Level + Justification

Confidence Level: Medium-High

Justification:

- Hash-1 and Hash-2 have strong multi-vendor detections on VirusTotal and clear behavioral evidence (sandbox flags as Stealer/Trojan/Evader + AgentTesla attribution via OTX pulses).
- The related infrastructure (103.101.85.26) shows classic C2 panel characteristics, even though it is not yet widely blacklisted.
- Confidence is not "High" because some contacted IPs belonged to legitimate CDNs (Akamai/Microsoft) and OTX had limited pulses on Hash-1.
  Overall, the combination of AV detections, behavioral analysis, and community intelligence supports a solid Medium-High confidence assessment.

Conclusion

This lab successfully demonstrated a complete junior CTI workflow: starting from raw hashes, performing multi-source enrichment (VirusTotal + OTX), pivoting to related infrastructure, mapping behaviors to MITRE ATT&CK, and producing actionable recommendations.

The exercise confirmed that free OSINT tools are highly effective for initial triage and threat assessment. The findings highlight typical information-stealer tactics and provide clear defensive actions. This approach mirrors real-world day-one analyst work and forms a solid foundation for more advanced threat intelligence investigations.